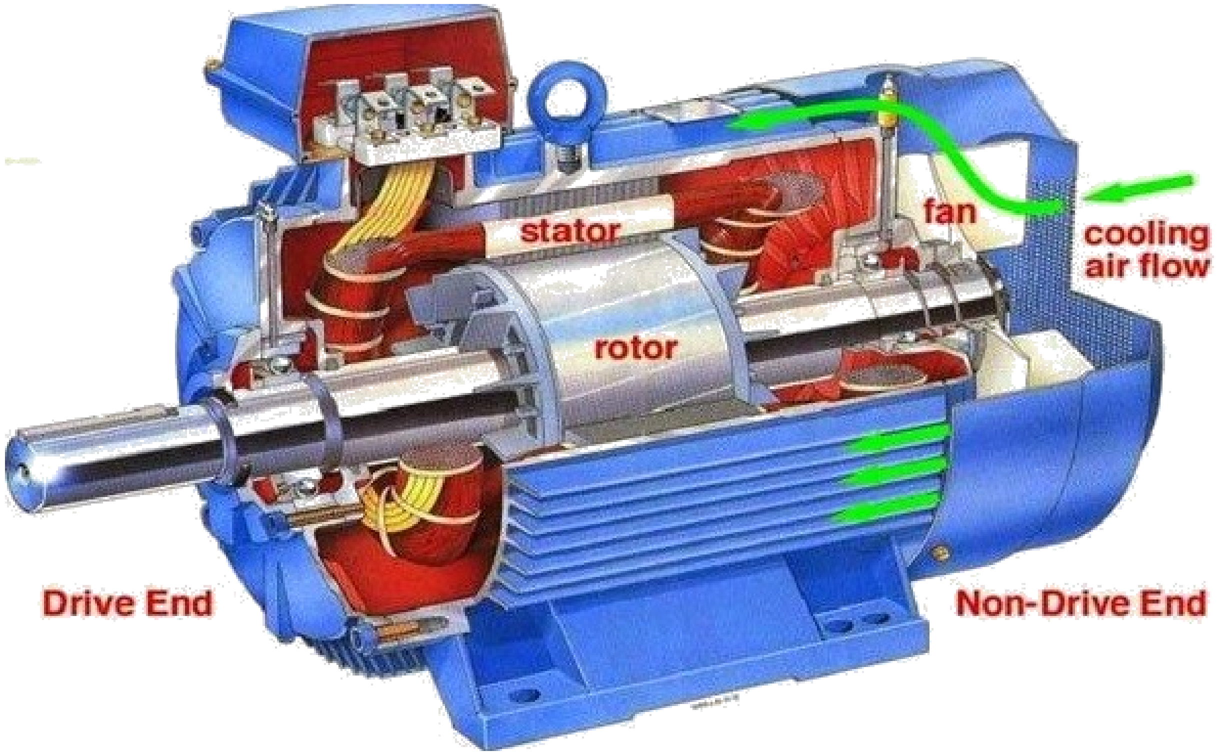
Section 2

INDUCTION MOTOR (IM)

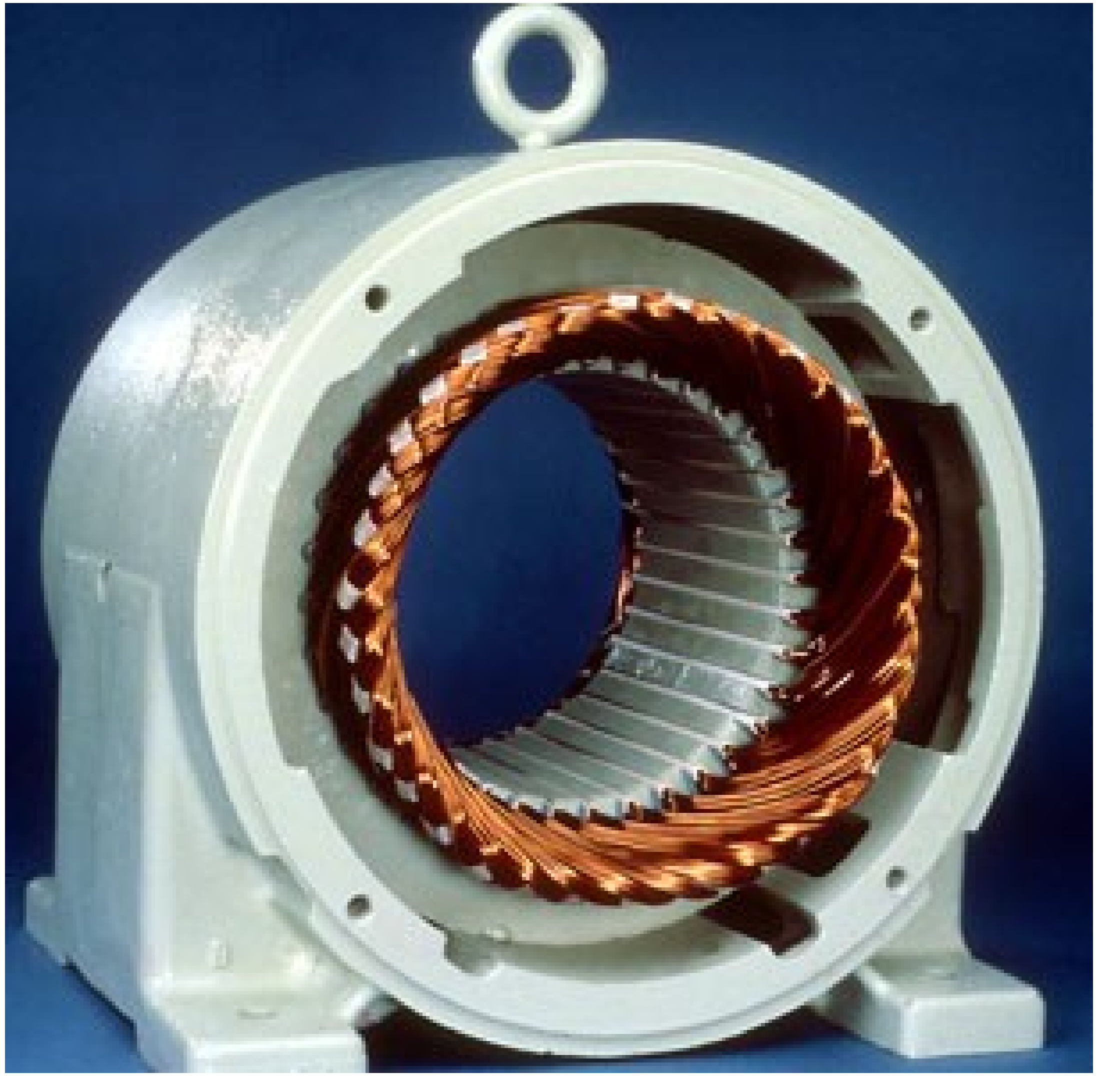
Introduction

Definition: Induction motor (IM) is an alternating-current motor in which torque is produced by the reaction between a varying magnetic field generated in the stator and the current induced in the coils of the rotor.



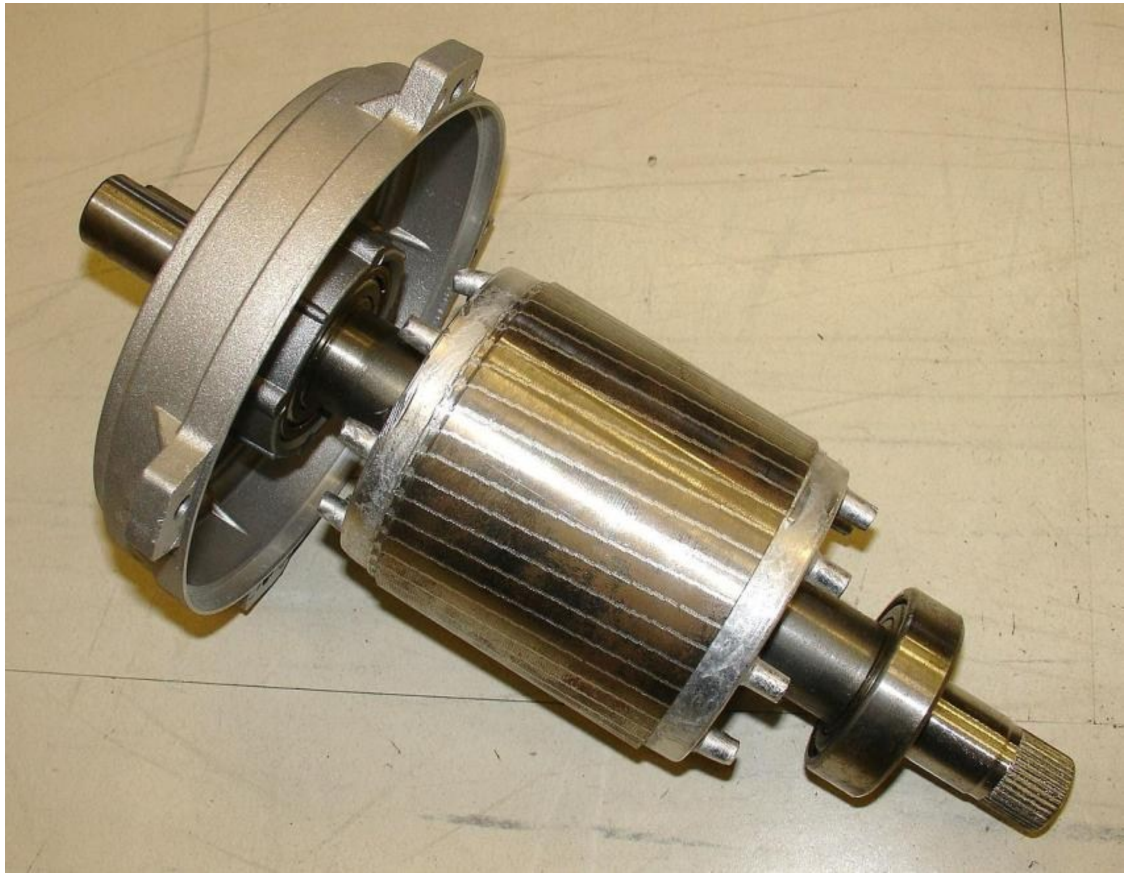
Construction of Induction motor

Stator of Induction Motor:



Two types of rotor:

Squirrel Cage



Wound rotor (with slip rings)

Working principle of Induction motor

Basic Induction Motor Concepts

- The speed of the magnetic field's rotation is given by:

$$\boxed{n_{sync} = \frac{120 f_e}{P}}$$

- The difference between synchronous speed and rotor speed:

$$\boxed{n_{slip} = n_{sync} - n_m}$$

- Slip is defined as:

$$s = \frac{n_{slip}}{n_{sync}} (\times 100\%)$$

$$\boxed{s = \frac{n_{sync} - n_m}{n_{sync}} (\times 100\%)}$$

$$n_m = \frac{30}{\pi} \omega_m$$

$$\boxed{s = \frac{\omega_{sync} - \omega_m}{\omega_{sync}} (\times 100\%)}$$

- It is possible to express the mechanical speed of the rotor shaft in term of synchronous speed and slip:

$$\boxed{n_m = (1 - s)n_{sync}}$$

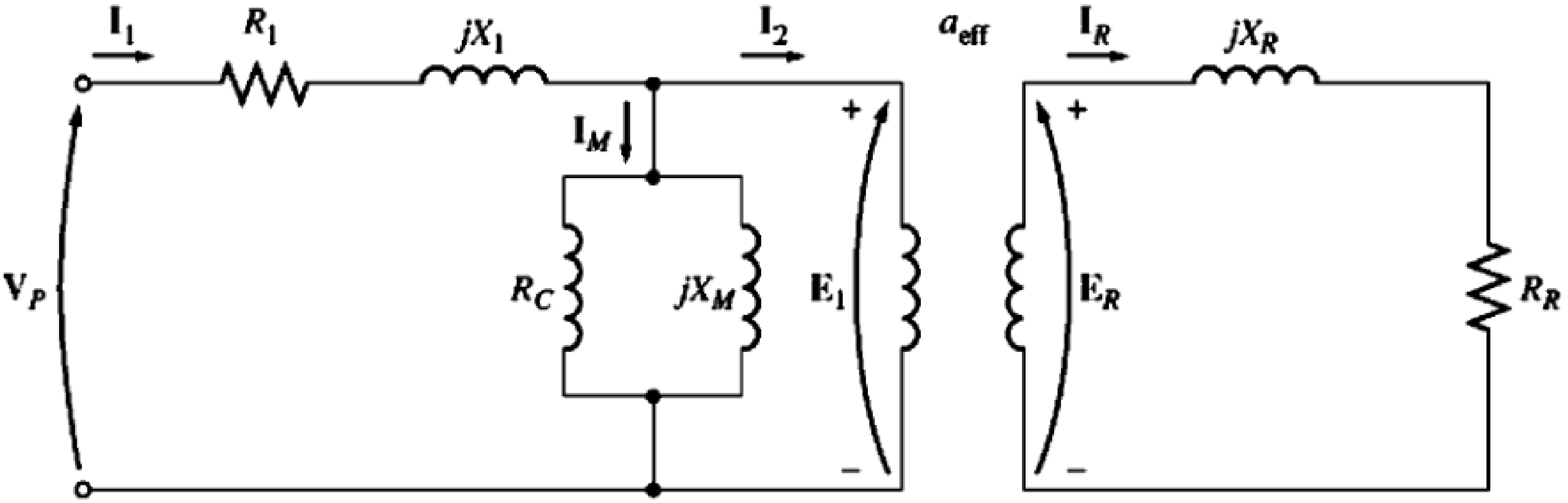
$$\boxed{\omega_m = (1 - s)\omega_{sync}}$$

- The rotor frequency can be expressed as:

$$\boxed{f_r = sf_e}$$

$$\boxed{f_r = \frac{P}{120}(n_{sync} - n_m)}$$

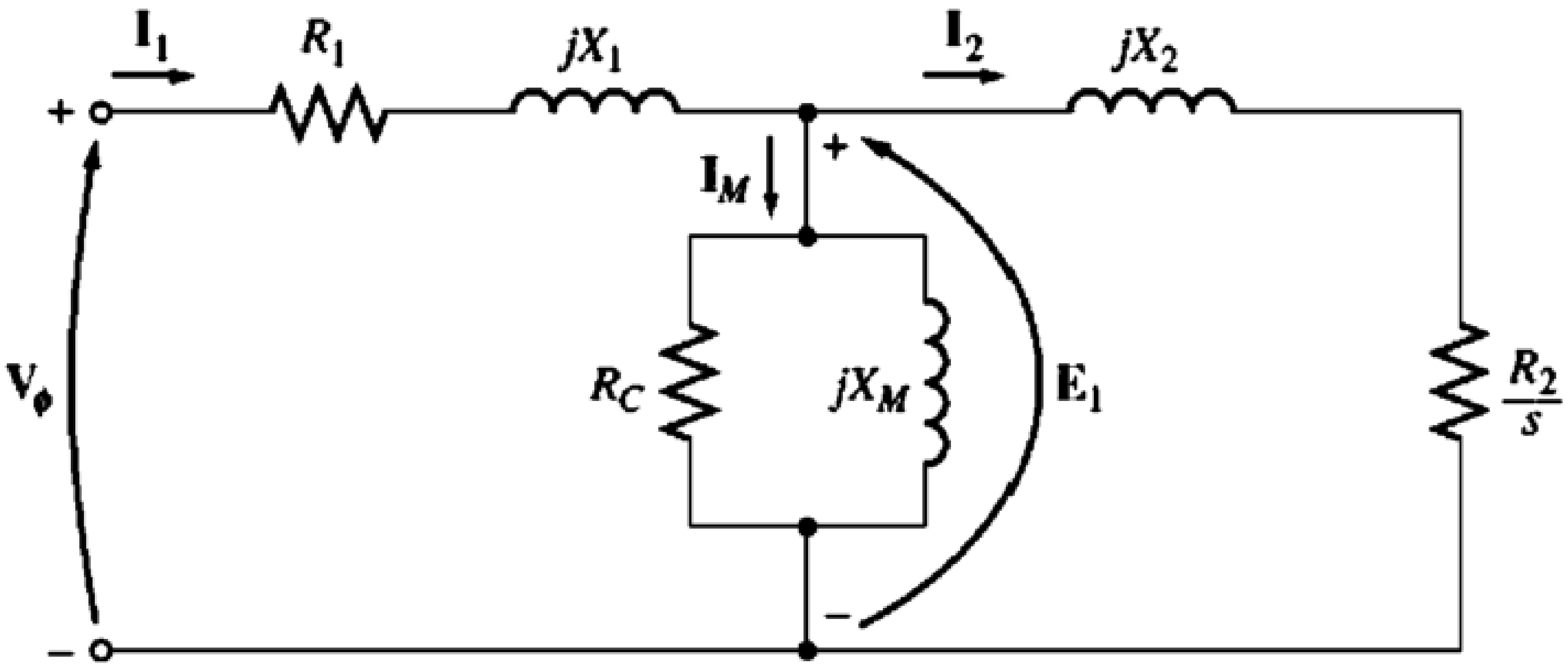
The Equivalent Circuit



$$X_R = \omega_r L_R = 2\pi f_r L_R$$

$$\mathbf{I}_R = \frac{\mathbf{E}_R}{R_R + jX_R}$$

The Equivalent Circuit



$$R_2 = a_{\text{eff}}^2 R_R$$

$$X_2 = a_{\text{eff}}^2 X_{R0}$$

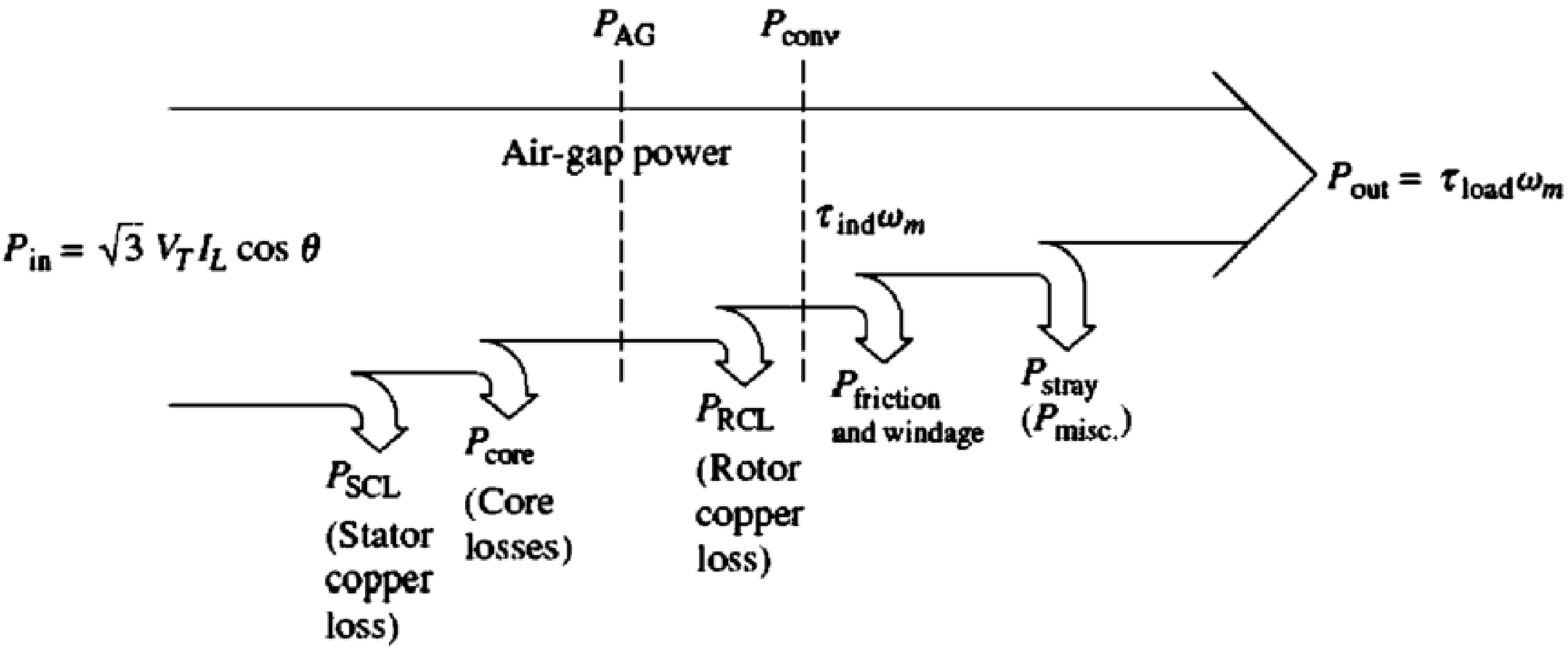
$$\mathbf{E}_1 = \mathbf{E}'_R = a_{\text{eff}} \mathbf{E}_{R0}$$

E_{R0} : the induced rotor voltage at locked-rotor

X_{R0} : is the blocked-rotor reactance

a_{eff} : the ratio of the conductors per phase on the stator to the conductors per phase on the rotor

Power Flow Diagram



$$P_{SCL} = 3I_1^2 R_1$$

$$p_{core} = 3E_1^2 / R_C$$

$$P_{AG} = 3I_2^2 (R_2 / S)$$

$$P_{RCL} = 3I_2^2 R_2 = sP_{AG}$$

$$P_{conv} = P_{AG} - P_{RCL} = 3I_2^2 [R_2 (1-S)/S] = (1-s) P_{AG}$$

- The induced torque is given by the equation:

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m}$$

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}}$$

- Load torque:

$$\tau_{load} = \frac{P_{out}}{\omega_m}$$

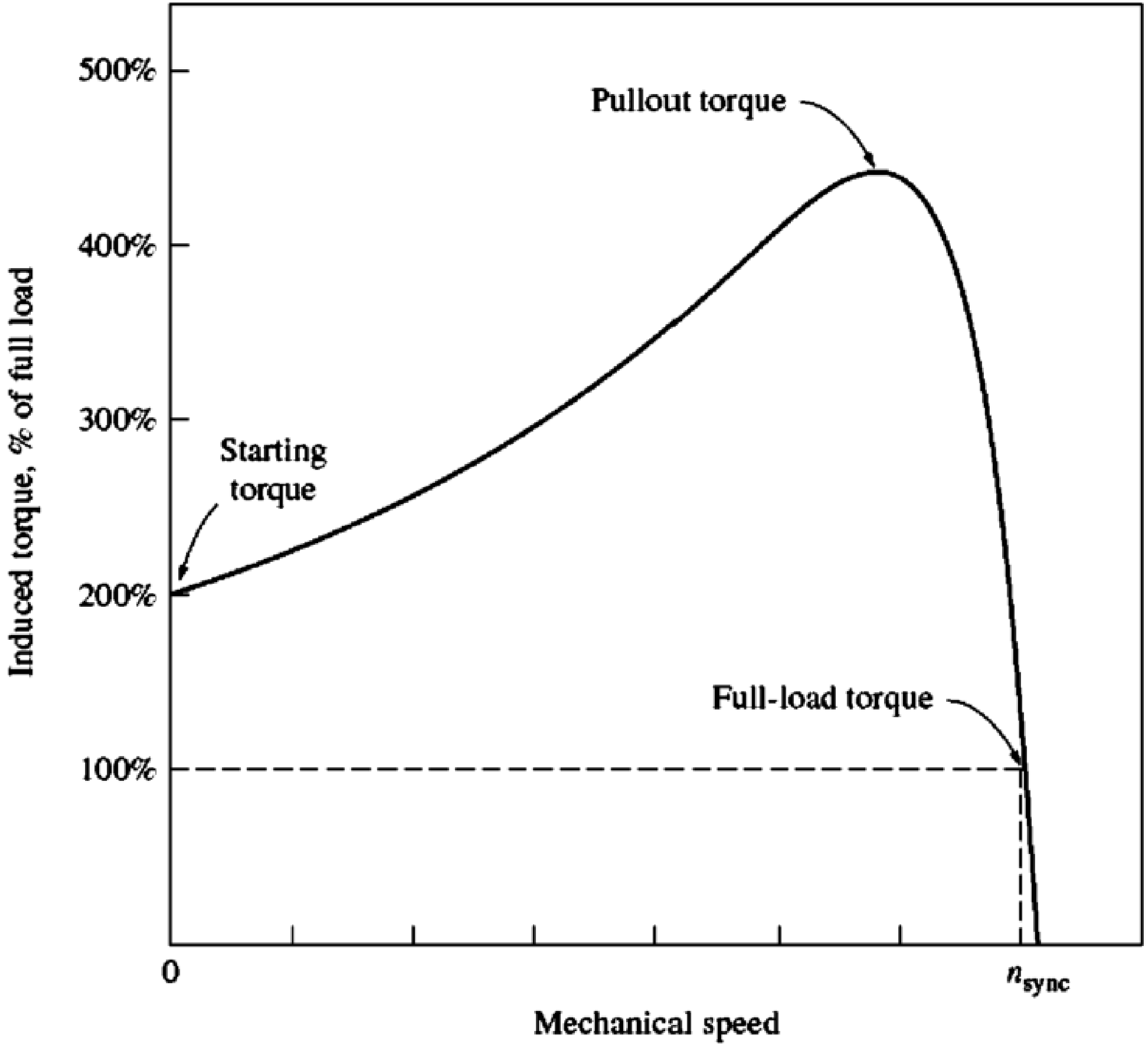
Torque Speed Characteristic

$$\boxed{\tau_{\text{ind}} = \frac{3V_{\text{TH}}^2 R_2/s}{\omega_{\text{sync}}[(R_{\text{TH}} + R_2/s)^2 + (X_{\text{TH}} + X_2)^2]}}$$

$$V_{\text{TH}} \approx V_\phi \frac{X_M}{X_1 + X_M}$$

$$R_{\text{TH}} \approx R_1 \left(\frac{X_M}{X_1 + X_M} \right)^2$$

$$X_{\text{TH}} \approx X_1$$



- The maximum or pullout torque and slip are:

$$\tau_{max} = \frac{3V_{TH}^2}{2\omega_{sync}\left[R_{TH} + \sqrt{R_{TH}^2 + (X_{TH} + X_2)^2}\right]}$$

$$s_{max} = \frac{R_2}{\sqrt{R_{TH}^2 + (X_{TH} + X_2)^2}}$$

- Starting torque is (s = 1):

$$\tau_{start} = \frac{3V_{TH}^2 R_2}{\omega_{sync}\left[(R_{TH} + R_2)^2 + (X_{TH} + X_2)^2\right]}$$

Example 1:

A 480-V, 60-Hz, 50-hp, three-phase induction motor is drawing 60 A at 0.85 PF lagging. The stator copper losses are 2kW, and the rotor copper losses are 700 W. The friction and windage losses are 600 W, the core losses are 1800 W, and the stray losses are negligible. Find the following quantities:

(a) The air-gap power P_{AG}

(b) The power converted P_{conv}

(c) The output power P_{out}

(d) The efficiency of the motor η

Example 2:

A 208-V, 10-hp, four-pole, 60-Hz, Y-connected induction motor has a full-load slip of 5 percent.

(a) What is the synchronous speed of this motor?

(b) What is the rotor speed of this motor at the rated load?

(c) What is the rotor frequency of this motor at the rated load?

(d) What is the shaft torque of this motor at the rated load?